

CE 601: Geo-Informatics

Course Number: CE 601

Course Name: Geo-Informatics

L-T-P-C: 2-0-2-3

Pre-requisite: CE 508 - Photogeology and Photogrammetry; CE 501 - Remote Sensing

Intended for: B.Tech. 4th year/M.S./M.Tech./Ph.D.

Distribution: Elective

Approval: 9th Senate

Course Contents

Module I: Introduction, concepts and terminology, GIS packages & components of GIS, difference between image processing system and GIS. [5 Lectures]

Module II: Elements of Image interpretation, Image formats, Raster and vector data, Data acquisition through scanners and digitizers, methods of digitization. [5 Lectures]

Digital Data: spatial & non-spatial; preprocessing, spatial referencing and positioning, rectification and registration, interpolation. [10 Lectures]

Database structure: Hierarchical data, network systems, relational database, and data management in GIS. [3 Lectures]

Module V: Data visualization, data manipulation, overlay, buffering, interpolation, query analysis, mathematical operations on data. [9 Lectures]

Module VI: Project based application of GIS for various natural resources mapping & monitoring and for engineering applications. [10 Lectures]

Text Books:

1. Kang-Tsung Chang, **Introduction to geographic information systems**, 8th Edition, McGraw-Hill Education, 2015.
2. JR Jensen, and RR Jensen, **Geographic information systems : Exercise workbook**, Pearson College Division, 2013.
3. M A. Gomarasca, **Basics of Geomatics**, Springer, 2009.
4. P A. Longley, M Goodchild, D J. Maguire, D W. Rhind, **Geographic Information Systems and Science**, 3rd Edition, John Wiley, 2010.

Reference Books:

1. P.A. Burrough, and R.A., McDonnell, **Principles of Geographic Information for Land Resources Assessment**, Oxford University Press, USA, 1998.
2. M.N., DeMers, **Fundamentals of Geographic Information System**, 3rd Edition, John Wiley, 2008.
3. M. Neteler, and H. Mitasova, **Open source GIS : GRASS GIS approach**, Springer, 2008.
4. R S. Lunetta, J G. Lyon, **Remote Sensing and GIS Accuracy Assessment**, CRC Press, 2005.